

ML Model Deployment with Docker & FastAPI

This document explains the end-to-end workflow for deploying machine learning models using FastAPI and Docker. It covers model preprocessing, serialization, API endpoint creation, containerization, monitoring, and scaling.

Key Concepts:

- Preprocessing: Data cleaning, feature engineering, model training
- Model Packaging: Pickle/Joblib serialization
- FastAPI Inference API: /predict endpoint, async handling
- Docker Deployment: Creating lightweight images for production
- Scaling: Load balancing containers, Kubernetes-ready structure
- Monitoring: Logging, metrics, auto-reload

Frameworks Used:

- FastAPI
- Docker
- Scikit-learn